An Obvious Theorem, Proved Twice

On the convergence of r^n to 0 for $|r| < 1$

This document was generated by a large language model (LLM) and may contain errors.

Abstract

The assertion that $r^n \to 0$ whenever $|r| < 1$ is among the first facts a student meets about sequences, and it is usually treated as self-evident: repeated multiplication by something small ought to produce something small. This note takes the statement seriously. We isolate exactly which axioms of the real number system the result depends on, prove it twice by genuinely different mechanisms, and exhibit an ordered field in which the statement is *false*. The purpose is not the theorem, which is not in doubt, but the discipline: to display what it looks like to discharge every obligation in an argument whose conclusion nobody disputes.

Contents

1 Statement and standing conventions

Theorem 1.1. *Let $r \in \mathbb{C}$ with $|r| < 1$. Then the sequence $(r^n)_{n\in\mathbb{N}}$ converges to 0; that is,*

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \forall n \geq N : \quad |r^n - 0| < \varepsilon.$$

We write $\mathbb{N} = \{0, 1, 2, \dots\}$. Throughout, $\mathbb{R}$ denotes a complete ordered field, and $|\cdot|$ denotes the usual absolute value on $\mathbb{R}$ or the modulus on $\mathbb{C}$, as context requires. The only properties of $\mathbb{R}$ that will be invoked are the ordered-field axioms together with the following.

Definition 1.2 (Least upper bound axiom)**.** Every nonempty subset of $\mathbb{R}$ that is bounded above possesses a least upper bound (supremum) in $\mathbb{R}$.

Every appeal to "completeness" below is an appeal to Definition 1.2, directly or through one of its consequences. This bookkeeping is the point of the essay: the theorem is not a fact about arithmetic, it is a fact about the completeness of $\mathbb{R}$, and Section 5 makes that concrete.

2 Reduction to the real, positive case

Lemma 2.1 (Multiplicativity of the modulus on powers)**.** *For all $r \in \mathbb{C}$ and all $n \in \mathbb{N}$ we have $|r^n| = |r|^n$.*

Proof. Induction on n. For $n = 0$, both sides equal 1, since $r^0 = 1$ and $|1| = 1$. Suppose $|r^n| = |r|^n$ for some $n \in \mathbb{N}$. Then, using multiplicativity of the modulus,

$$|r^{n+1}| = |r^n \cdot r| = |r^n| \cdot |r| = |r|^n \cdot |r| = |r|^{n+1}.$$

By induction the identity holds for all $n \in \mathbb{N}$. $\square$

Proposition 2.2 (Reduction)**.** *Theorem 1.1 follows from the special case in which r is real and $0 < r < 1$.*

Proof. Set $p := |r|$, so $p \in [0, 1)$ by hypothesis. By Lemma 2.1,

$$|r^n - 0| = |r^n| = p^n = |p^n - 0| \qquad \text{for every } n \in \mathbb{N},$$

so the statement "$r^n \to 0$" and the statement "$p^n \to 0$" unfold to the identical quantified assertion, and each holds if and only if the other does.

If $p = 0$, then $r = 0$ and $r^n = 0$ for all $n \geq 1$; the definition of convergence is satisfied with $N = 1$ for every $\varepsilon > 0$, since $|r^n - 0| = 0 < \varepsilon$. Otherwise $p \in (0, 1)$, which is the special case assumed available. $\square$

For the remainder of the essay we fix $p \in \mathbb{R}$ with $0 < p < 1$ and prove that $p^n \to 0$.

Lemma 2.3 (Positivity of the powers)**.** *$p^n > 0$ for every $n \in \mathbb{N}$.*

Proof. Induction. $p^0 = 1 > 0$. If $p^n > 0$, then $p^{n+1} = p^n \cdot p$ is a product of two strictly positive elements of an ordered field, hence strictly positive. $\square$

3 First proof: monotone convergence and the fixed-point equation

The strategy is to separate two questions that are easy to conflate: *whether* the limit exists, and *what* its value is. Completeness answers the first; an algebraic identity answers the second.

3.1 The tools

Lemma 3.1 (Monotone convergence theorem, decreasing form)**.** *Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of real numbers that is nonincreasing ($a_{n+1} \leq a_n$ for all n) and bounded below. Then (a_n) converges, and*

$$\lim_{n\to\infty} a_n = \inf\{a_n : n \in \mathbb{N}\}.$$

Proof. Let $S = \{a_n : n \in \mathbb{N}\}$, a nonempty set bounded below. Then $-S$ is nonempty and bounded above, so by Definition 1.2 it has a supremum; setting $L := -\sup(-S)$ gives the infimum of S. We claim $a_n \to L$.

Let $\varepsilon > 0$. Since $L + \varepsilon > L$ and L is the *greatest* lower bound of S, the number $L + \varepsilon$ is not a lower bound of S; hence there exists $N \in \mathbb{N}$ with $a_N < L + \varepsilon$. For any $n \geq N$, monotonicity gives $a_n \leq a_N$ (a trivial induction on $n - N$), while L being a lower bound gives $L \leq a_n$. Therefore

$$L - \varepsilon < L \leq a_n \leq a_N < L + \varepsilon,$$

so $|a_n - L| < \varepsilon$. As $\varepsilon > 0$ was arbitrary, $a_n \to L$. □

Lemma 3.2 (Shift invariance of limits). *If $a_n \to L$, then $a_{n+1} \to L$.*

Proof. Let $\varepsilon > 0$ and choose N with $|a_n - L| < \varepsilon$ for all $n \geq N$. If $n \geq N$ then $n + 1 \geq N$, so $|a_{n+1} - L| < \varepsilon$. □

Lemma 3.3 (Scalar multiples). *If $a_n \to L$ and $c \in \mathbb{R}$, then $c\,a_n \to cL$.*

Proof. If $c = 0$ the claim is immediate. Otherwise let $\varepsilon > 0$ and choose N with $|a_n - L| < \varepsilon/|c|$ for all $n \geq N$; then $|ca_n - cL| = |c|\,|a_n - L| < \varepsilon$ for such n. □

Lemma 3.4 (Uniqueness of limits). *A sequence of real numbers has at most one limit.*

Proof. Suppose $a_n \to L$ and $a_n \to M$ with $L \neq M$. Put $\varepsilon := |L - M|/2 > 0$ and choose N_1, N_2 witnessing convergence to L and to M respectively for this ε. For $n \geq \max(N_1, N_2)$ the triangle inequality gives

$$|L - M| \leq |L - a_n| + |a_n - M| < \varepsilon + \varepsilon = |L - M|,$$

a contradiction. □

3.2 The proof

First proof of Theorem 1.1. By Proposition 2.2 it suffices to show $p^n \to 0$ for $0 < p < 1$.

Step 1: the sequence is strictly decreasing and bounded below. By Lemma 2.3, $p^n > 0$ for every n, so 0 is a lower bound. Moreover, multiplying the strict inequality $p < 1$ by the strictly positive number p^n preserves it, giving

$$p^{n+1} = p^n \cdot p < p^n \cdot 1 = p^n.$$

Step 2: the sequence converges. By Step 1 and Lemma 3.1, there exists $L \in \mathbb{R}$ with $p^n \to L$. Since 0 is a lower bound for the terms and L is their infimum, $L \geq 0$.

Step 3: the limit satisfies $L = pL$. Apply Lemma 3.2 to the convergent sequence (p^n): the shifted sequence (p^{n+1}) converges to L as well. Apply Lemma 3.3 with $c = p$ to the same sequence: since $p^{n+1} = p \cdot p^n$, the sequence (p^{n+1}) converges to pL. By Lemma 3.4 a sequence cannot have two distinct limits, so

$$L = pL.$$

Step 4: conclude. Rearranging, $L(1-p) = 0$. In a field there are no zero divisors, and $1 - p \neq 0$ because $p < 1$. Hence $L = 0$, i.e. $p^n \to 0$. □

Remark 3.5 (The order of Steps 2 and 3 is not negotiable). The identity $L = pL$ is meaningless until L has been shown to exist. This is the standard error in this proof, and it is not a pedantic one: the manipulation "$L = \lim q^{n+1} = q \lim q^n = qL$, hence $L(1-q) = 0$, hence $L = 0$" applies verbatim to $q = 2$ and yields the false conclusion that $2^n \to 0$. What fails for $q = 2$ is precisely Step 2, and nothing else. An argument whose validity turns entirely on a step one is tempted to omit is an argument worth writing out.

Remark 3.6 (What Step 3 really is). Step 3 says the limit must be a fixed point of the map $x \mapsto px$. That map has 0 as its unique fixed point exactly because $p \neq 1$. The first proof is thus a rigidity argument: completeness produces a limit, and the algebra of the recursion leaves it no room to be anything but 0.

4 Second proof: Bernoulli's inequality and the Archimedean property

The first proof establishes convergence without ever producing an N explicitly. The second proof does the opposite: it constructs an explicit bound on p^n and reads N off from it. It uses completeness only through the Archimedean property.

4.1 The tools

Lemma 4.1 (Bernoulli's inequality). *Let $h \in \mathbb{R}$ with $h \geq -1$. Then $(1+h)^n \geq 1 + nh$ for every $n \in \mathbb{N}$.*

Proof. Induction on n. For $n = 0$ both sides equal 1. Assume $(1+h)^n \geq 1+nh$. Since $1+h \geq 0$, multiplying this inequality by $1+h$ preserves its direction:

$$(1+h)^{n+1} = (1+h)^n(1+h) \geq (1+nh)(1+h) = 1 + (n+1)h + nh^2 \geq 1 + (n+1)h,$$

the last step because $nh^2 \geq 0$. This completes the induction. □

Lemma 4.2 (Archimedean property). *For every $x \in \mathbb{R}$ there exists $N \in \mathbb{N}$ with $N > x$.*

Proof. Suppose not. Then $\mathbb{N} \subseteq \mathbb{R}$ is nonempty and bounded above by x, so by Definition 1.2 it has a supremum s. Since $s - 1 < s$ and s is the least upper bound, $s - 1$ is not an upper bound: there is $m \in \mathbb{N}$ with $m > s - 1$. But then $m + 1 \in \mathbb{N}$ and $m + 1 > s$, contradicting that s is an upper bound for $\mathbb{N}$. □

4.2 The proof

Second proof of Theorem 1.1. Again it suffices to treat p with $0 < p < 1$.

Step 1: a change of variable. Define

$$h := \frac{1}{p} - 1 = \frac{1-p}{p}.$$

Since $0 < p < 1$, both numerator and denominator are strictly positive, so $h > 0$; and by construction $1 + h = 1/p$, that is,

$$p = \frac{1}{1+h}, \qquad h > 0.$$

Step 2: an explicit upper bound for p^n. Raising the previous identity to the n-th power and applying Lemma 4.1,

$$p^n = \frac{1}{(1+h)^n} \le \frac{1}{1+nh} \qquad \text{for all } n \in \mathbb{N},$$

where the inequality is obtained from $(1+h)^n \ge 1+nh > 0$ by taking reciprocals, which reverses inequalities between strictly positive quantities. Combining with Lemma 2.3,

$$0 < p^n \le \frac{1}{1+nh} < \frac{1}{nh} \qquad \text{for all } n \ge 1. \tag{1}$$

Step 3: choose N. Let $\varepsilon > 0$ be given. By Lemma 4.2 applied to the real number $1/(h\varepsilon)$, there exists $N \in \mathbb{N}$ with

$$N > \frac{1}{h\varepsilon}.$$

Note $N \ge 1$, since $1/(h\varepsilon) > 0$. For any $n \ge N$ we have $nh \ge Nh > 1/\varepsilon > 0$, hence $1/(nh) < \varepsilon$, and so by (1),

$$|p^n - 0| = p^n < \frac{1}{nh} < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $p^n \to 0$. $\square$

Remark 4.3 (The two proofs are genuinely different)*.* The first proof is non-constructive in an essential way: it produces the limit as a supremum supplied by Definition 1.2 and never exhibits, for a given ε, any particular index beyond which the terms are small. The second proof is quantitative. It yields the explicit and reusable estimate

$$p^n \le \frac{1}{1 + n\left(\frac{1}{p} - 1\right)} = O(n^{-1}),$$

which is a crude bound—the truth is geometric decay—but a bound one can sharpen. Replacing Bernoulli's inequality by the stronger binomial estimate $(1+h)^n \ge \binom{n}{2}h^2$, valid for $h > 0$ and $n \ge 2$, gives $p^n = O(n^{-2})$ and hence the strictly stronger conclusion $np^n \to 0$, which does not follow from the first proof without further work.

5 Why the theorem is not obvious: a countermodel

An argument's necessity is best demonstrated by removing a hypothesis and watching the conclusion fail. Both proofs above used completeness: the first through the monotone convergence theorem, the second through the Archimedean property. Neither is a consequence of the ordered-field axioms alone, and without them the theorem is false.

Consider the field $\mathbb{R}(t)$ of rational functions with real coefficients, ordered by declaring $f > 0$ if and only if $f(x) > 0$ for all sufficiently large real x. One verifies without difficulty that this is a total order compatible with the field operations, so $\mathbb{R}(t)$ is an ordered field containing $\mathbb{R}$ as an ordered subfield. In it, the element t satisfies $t > n$ for every $n \in \mathbb{N}$; consequently $\varepsilon_0 := 1/t$ satisfies

$$0 < \varepsilon_0 < \frac{1}{n} \qquad \text{for every } n \ge 1.$$

This field is therefore non-Archimedean: Lemma 4.2 fails, and with it Definition 1.2.

Now take $p = 1/2$, an element of $\mathbb{R}(t)$ satisfying $0 < p < 1$. For every $n \in \mathbb{N}$ the quantity 2^{-n} is an ordinary positive real number, while ε_0 is strictly smaller than every positive real; hence

$$p^n = 2^{-n} > \varepsilon_0 \qquad \text{for every } n \in \mathbb{N}.$$

Thus the decreasing sequence (p^n) is bounded below by the strictly positive element ε_0, and no matter how the notion of limit is formulated in the order topology of $\mathbb{R}(t)$, the sequence does not converge to 0: the neighbourhood $(-\varepsilon_0, \varepsilon_0)$ of 0 contains no term of the sequence at all. In fact (p^n) has no limit whatsoever, since the set of its lower bounds—which includes every positive infinitesimal—has no greatest element.

The moral is exact. The arithmetic of p^n is identical in $\mathbb{R}$ and in $\mathbb{R}(t)$: the same recursion, the same monotonicity, the same positivity. Everything that "obviously" makes the powers shrink to nothing is present in the countermodel. What is absent is completeness. The theorem is therefore not a statement about multiplication making things smaller; it is a statement about $\mathbb{R}$ having no gaps for the sequence to hide in.

6 Summary of dependencies

First proof	**Second proof**
Ordered-field axioms	Ordered-field axioms
Least upper bound axiom	Least upper bound axiom
via monotone convergence (Lem. 3.1)	via the Archimedean property (Lem. 4.2)
Uniqueness of limits (Lem. 3.4)	Bernoulli's inequality (Lem. 4.1)
Algebra of limits (Lem. 3.2, 3.3)	Induction
Induction	
Non-constructive	Constructive: $N > \dfrac{p}{(1-p)\varepsilon}$

Neither proof appeals to logarithms, exponentials, power series, the ratio test, or any part of the theory of geometric series. This is deliberate. All of those tools are downstream of Theorem 1.1: the convergence of $\sum r^n$ for $|r| < 1$, the radius-of-convergence machinery, and the analytic construction of log each rest on the fact proved here. A proof of Theorem 1.1 that invokes them is not a proof but a circle, however impressive its vocabulary. The two arguments above are, in a precise sense, as deep as this statement is allowed to go.